Week 10 Progress Report

Predictive Modeling and What-If Analysis

Dynamic Hotel Room Pricing and Overbooking Optimization DSS

Master's Research Project (MS Analytics) — Group 11

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1. Key Performance Indicators and Justification

We selected two KPIs that directly shape the hotel revenue manager's nightly decisions.

KPI 1 — Booking cancellation probability

What it measures: the probability that a single reservation is cancelled before arrival.

Relevance: cancellations are the main source of uncertainty for the overbooking decision. A reliable per-booking probability lets the hotel sell slightly above capacity without walking guests, protecting occupancy revenue (operational and financial performance) and guest goodwill (reputation).

KPI 2 — Expected nightly occupancy rate

What it measures: the expected share of rooms occupied on a future stay-date (0–100%).

Relevance: expected occupancy is the demand signal behind pricing and overbooking aggressiveness. High expected occupancy supports firmer rates and tighter inventory; low occupancy supports discounts and caution. This connects directly to RevPAR and profitability.

2. Predictive Models

KPI 1 — Gradient Boosting Classifier

Predictors and why: lead time, ADR, party size (adults, children), previous cancellations, repeat-guest flag, special requests, total nights, hotel, market segment, booking channel, and deposit type. These are all known when a reservation is created, so the model uses no post-decision information (no leakage).

Algorithm and rationale: Gradient Boosting Classifier. Cancellation depends on non-linear interactions — for example, a long lead time combined with a non-refundable deposit behaves very differently from either alone. Gradient boosting captures these interactions and pairs cleanly with SHAP TreeExplainer for exact per-booking explanations. Validated on an out-of-time split: ROC-AUC ≈ 0.88, Brier ≈ 0.12.

KPI 2 — Gradient Boosting Regressor

Predictors and why: month, season, week number, day of month, weekend flag, year, and hotel — the calendar position of a future stay-date, all available in advance.

Algorithm and rationale: Gradient Boosting Regressor. Demand is seasonal and non-linear across the calendar; gradient boosting captures that shape and supports SHAP. Out-of-time R^2 ≈ 0.18, MAE ≈ 10 occupancy points. The modest R^2 reflects real night-to-night demand noise, so the model is presented as a guide used alongside manager judgment, not a guarantee.

3. Expanded Application Functionality

A new screen, Predictive Models & What-If, was added to the existing Streamlit application. For each KPI it provides all four required capabilities:

Requirement	How the app delivers it
(a) Predictor importance and contributions	Overall importance = mean |SHAP| across many cases (whole model). Per-prediction contributions = SHAP values for the single what-if case. The two are shown in separate, clearly labeled sections.
(b) Contextual interpretation	Each contribution is read as direction (sign), magnitude (size), and practical meaning, with an explicit statement that these are associations, not causation.
(c) What-if analysis	Users edit predictor values and see the scenario prediction next to a fixed baseline prediction, with the change in points. An input change may or may not move the prediction.
(d) Auto-generated summary	After each what-if, the app writes a summary of the scenario inputs, the updated prediction, the change from baseline, and the implications and limitations (model-based scenario, not a guaranteed real-world effect).

4. Work Accomplished to Date

- Weeks 1–5: defined the decision problem, built and validated a fully synthetic relational dataset calibrated to the Antonio et al. (2019) hotel-booking data (22/22 validation checks pass).
- Week 6: low-fidelity UI blueprint and decision logic.
- Week 9: first working Streamlit prototype — Hotel Overview and Overbooking Decision Support screens, with a cancellation model and Monte Carlo buffer optimization.
- Week 10 (this deliverable): added two predictive KPIs with Gradient Boosting models, SHAP-based global importance and per-prediction contributions, interactive what-if with baseline comparison, and auto-generated scenario summaries.

5. Group Member Contributions

Member	Contribution
Chao-Ting Chueh	Predictive modeling (Gradient Boosting for both KPIs), SHAP explainability, what-if logic, and model validation.
Bilal Syed	Streamlit application screens, data pipeline integration, auto-summary and interpretation text, progress report, and demonstration video.

Note: update the member names and the split above to match your team before submitting.

6. Important Limitations

- All data is synthetic (calibrated to real benchmarks); results are model-based scenarios, not live hotel decisions.
- SHAP contributions show association learned from historical data, not cause and effect.
- The occupancy model explains the seasonal component of demand; substantial night-level variation remains.
- The 70% contribution-margin assumption used in the overbooking screen is not yet sourced to an industry figure.